cp_10f

1

Hollow Pyramid | 35/35

Easy

Given a number `N` , write a program to print a **Hollow Pyramid** of `N` rows using stars (`*`).

```
    *
   * *
  *   *
 * * * *
```

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as a **Hollow Pyramid** shown in the sample output.

Explanation

For example, if the given number is `N = 5` ,

- Print stars (`*`) on **5** rows as shown in the given image.

Note

In the given image,

- The Empty box indicates a space (" ").
- The Star in the box indicates a star ("*").

The output should be,

```
    *
   * *
  *   *
 *     *
* * * * *
```

code:

```
n=int(input())
n1=n
for i in range(1,n+1):
    if i==1:
        row=" "*(n-1)+"* "
        print(row)
    elif i==n1:
        row="* "*n1
        print(row)
    else:
        left_space=" "*(n-2)
        hollow_space="  "*(i-2)
        row=left_space+"* "+hollow_space+"* "
        print(row)
        n=n-1
```

2

Hollow Right Angled Triangle - 4 | 40/40

Solved

Easy

Given a number `N` , write a program to print a **Hollow Right Angled Triangle** of `N` rows using pluses (`+`) and hashes (`#`).
The first line contains hashes (`#`) and the next `N - 1` lines contain pluses (`+`)

```
# # # # # #
+       +
+     +
+   +
+ +
+
```

Note

There is a space after every plus (`+`) and every hash (`#`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing hashes (`#`) and pluses (`+`) as a **Hollow Right Angled Triangle** as shown in the sample output.

Explanation

For example, if the given number is `N = 6` ,

- Print Hashes (`#`) and Pluses (`+`) on **6** rows as shown in the given image.
  - The first row contains Hashes (`#`).
  - The next **5** (`6 - 1`) rows contain Pluses (`+`).

code:

```
n=int(input())
n1=n
for i in range(1,n+1):
    if i==1:
        row="# "*n
        print(row)
    elif i==n1:
        print("+")
    else:
        spacing="  "*(n-2)
        row="+ "+spacing+"+ "
```

```
        print(row)
    n=n-1
```

3

Hollow Pyramid - 2 | 40/40

Easy

Given a number `N` , write a program to print a **Hollow Pyramid** of `2 * N - 1` rows using numbers.

```
1
2 2
3   3
4     4
3   3
2 2
1
```

Note

There is a space after every number (`1`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `2 * N - 1` rows containing numbers as a **Hollow Pyramid** shown in the sample output.

Explanation

For example, if the given number is `N = 4` ,

- Print Numbers on **7** rows (`2 * 4 - 1`) as shown in the given image.

code:

```
n=int(input())
for i in range(1,n+1):
    str_i=str(i)+" "
    if i==1:
        print(str_i)
    else:
```

```
        hollow_space=" "*(2*(i-2))
        row=str_i+hollow_space+str_i
        print(row)
n1=n-1
for i in range(1,n):
    if i==(n-1):
        row=str(1)+" "
        print(row)
    else:
        str_n1=str(n1)+" "
        space=" "*(2*(n1-2))
        row=str_n1+space+str_n1
        n1=n1-1
        print(row)
```

4

Hollow Right Angled Triangle - 3 | 30/30

So

Given a number `N` , write a program to print the **Hollow Right Angled Triangle** of `N` rows using stars (`*`).

```
* * * * *
*     *
*   *
* *
*
```

Note

There is a space after every star (`*`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as the **Hollow Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 5` ,

- Print the stars (`*`) on **5** rows as shown in the given image.

code:

```
n=int(input())
for i in range(1,n+1):
    if i==1:
        row="* "*n
        print(row)
    elif n==1:
        row="* "
        print(row)
    else:
        hollow_spaces=" "*(2*(n-2))
        row="* "+hollow_spaces+"* "
        print(row)
    n=n-1
```

5



Hollow Right Angled Triangle - 5 | 30/30

Solved

Medium

Given a number `N` , write a program to print the **Hollow Right Angled Triangle** of `N` rows using stars (`*`).

```
    *
   * *
  *   *
* * * *
```

Note

There is a space after every star (`*`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as the **Hollow Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 5` ,

- Print the stars (`*`) on **5** rows as shown in the given image.

code:

```
n=int(input())
n1=n
for i in range(1,n+1):
    if i==1:
        row="  "*(n-1)+"* "
        print(row)
    elif i==n1:
        row="* "*n1
        print(row)
    else:
        left_spaces=" "*(2*(n-2))
        star="* "
        hollow_spaces="  "*(i-2)
        row=left_spaces+star+hollow_spaces+star
```

```
        print(row)
        n=n-1
```

6

Hollow Right Angled Triangle - 6 | 30/30

Solved

Medium

Given a number `N` , write a program to print the **Hollow Inverted Right Angled Triangle** of `N` rows using stars (`*`).

```
* * * * *
  *     *
    *   *
    | * *
    |   *
```

Note

There is a space after every star (`*`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as the **Hollow Inverted Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 4` ,

- Print the stars (`*`) on **4** rows as shown in the given image.

code:

```
n=int(input())
n1=n
for i in range(1,n+1):
    if i==1:
        row="* "*n
        print(row)
    elif n==1:
        spaces=" "*(2*(n1-1))
        row=spaces+"* "
        print(row)
    else:
```

```
        left_space=" "*(2*(i-1))
        hollow_spaces=" "*(2*(n-2))
        star="* "
        row=left_space+star+hollow_spaces+star
        print(row)
    n=n-1
```

7

Inverted Hollow Pyramid | 35/35

Given a number `N` , write a program to print the **Hollow Inverted Full Pyramid** of `N` rows using stars (`*`).

```
* * * *
 *   *
  * *
   *
```

Input

The input will be a single line containing an integer representing `N` .

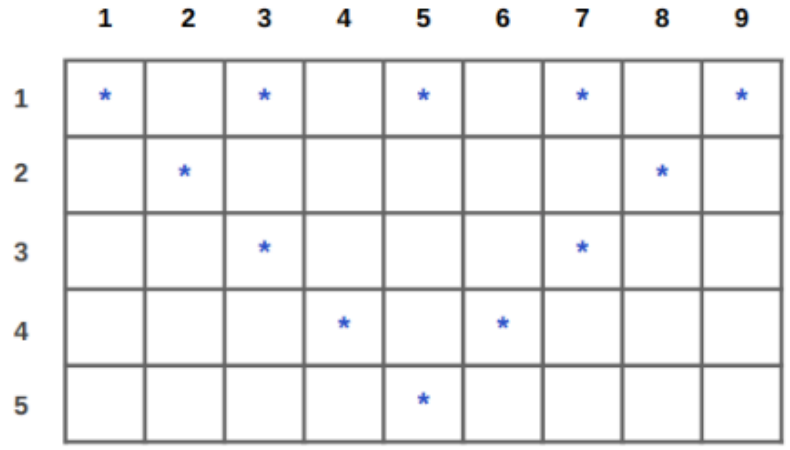
Output

The output should be `N` rows containing stars (`*`) as the **Hollow Inverted Full Pyramid** shown in the sample output.

Explanation

For example, if the given number is `N = 5` ,

- Print the stars (`*`) on **5** rows as shown in the given image.



code:

```
n=int(input())
n1=n-1
for i in range(1,n+1):
    if i==1:
        row="* "*n
        print(row)
    elif n==1:
        left_spaces=" "*n1
        row=left_spaces+"* "
        print(row)
    else:

        left_spaces=" "*(i-1)
        hollow_spaces=" "*(2*(n-2))
        star="* "
        row=left_spaces+star+hollow_spaces+star
        print(row)
    n=n-1
```

8

Hollow Diamond | 35/35

Given a number `N` , write a program to print the **Hollow Diamond** of `2 * N - 1` rows using stars (`*`).

```
    *
   * *
  *   *
 *     *
  *   *
   * *
    *
```

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `2 * N - 1` rows containing stars (`*`) as the **Hollow Diamond** shown in the sample output.

Explanation

For example, if the given number is `N = 5` ,

- Print the stars (`*`) on **9** rows (`2 * 5 - 1`) as shown in the given image.

code:

```
n=int(input())
n1=n-1
for i in range(1,n+1):
    if i==1:
        row=" "*(n-1)+"* "
        print(row)
    else:
       left_space=" "*(n-1)
       hollow_space=" "*(2*(i-2))
       star="* "
       row=left_space+star+hollow_space+star
       print(row)
    n=n-1
for i in range(1,n1+1):
    if n1==1:
        row=" "*i+"* "
        print(row)
    else:
```

```
        left_space=" "*i
        hollow_space=" "*(2*(n1-2))
        star="* "
        row=left_space+star+hollow_space+star
        print(row)
    n1=n1-1
```